Mehmet Sarp Bilgin

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Gameplay & systems programmer building original RPG systems on native Game Boy Advance C

Skills & Interests

Languages: C, C++, C#, Python, Java | Turkish, English
Frameworks & Tools: Godot, React, .NET 8, FastAPI, devkitARM, GNU Make, mGBA, Git, REST APIs
Leadership & Interests: Trip Leader, BU Outing Club | Piano, singing, pencil sketching, squash

Experience

Software Engineer Intern, SmartPulse | London, U.K. Jun 2026 - Jul 2026

- Built an authenticated .NET 8/C# REST service and React dashboard integrating EPİAŞ energy-market data; implemented TGT and cookie authentication, date-driven queries, plant search, filtering, and AG Grid views.
- Delivered transaction-summary, trade-value, and KGÜP endpoints; processed 38,127 transaction rows into 70 contract summaries with hourly volumes and weighted-average prices, and cached plant metadata.

AI Software Engineer Intern, OrbisLabs | Istanbul, Turkey May 2025 - Jul 2025

- Implemented Google's Agent-to-Agent protocol in Python/FastAPI; built an LLM orchestrator and Flights/Hotels agents that retrieved travel data, merged responses, and rendered answers in an internal web interface.
- Built a second orchestrator using the Model Context Protocol (MCP) for desktop automation; integrated it with Claude Desktop so natural-language instructions could edit and organize desktop content.

Software Engineer Intern, TaleWorlds Entertainment | Ankara, Turkey May 2024 - Jun 2024

- Built a procedural music-composition algorithm using chord/scale rules, Markov-inspired transitions, and complexity/dissonance weighting; modularized Note, Scale, and Chord components for reuse.
- Co-developed a playable monster-catching game in a three-day, two-person sprint; designed and implemented map progression, shops, capture mechanics, boss encounters, and approximately 120 catchable creatures.

Projects

Project Solarfly | Gameplay & Systems Programmer 2026 – Present

- Built an original RPG prototype on native Game Boy Advance C, including a deterministic 5v5 turn scheduler, effect-point-driven actions, targeting, and interacting status conditions.
- Structured the engine to separate hardware, battle logic, UI, data, and asset modules using C structs, state machines, and static data; implemented reusable move effects, resource costs, critical-hit rules, and battle menus.
- Engineered a GBA overworld under static, no-heap constraints with 16x16 metatile streaming, Q24.8 eight-direction movement, camera follow and clamping, foot-AABB collision with wall sliding, and event triggers.

PSP StudiGO | Audio & Systems Proggrammer 2026 – Present

- Built a digital audio workstation in native C for Sony PSP Go, supporting eight-track recording, clip arrangement, and piano-roll editing through a controller-driven interface.
- Engineered a 44.1 kHz, 48-voice audio engine with FM synthesis, SFZ sample playback, delay, and reverb; decoupled audio processing from UI rendering using a dedicated high-priority thread and lock-free command/event queues.
- Implemented sample-accurate sequencing, nondestructive clip editing, quantization, automation, and undo/redo; added versioned session save/load, Standard MIDI File export, and offline WAV mixdown.

Education

Boston University College of Arts and Sciences Boston, MA
Bachelor of Arts in Computer Science, GPA: 3.36 Sep 2023 - May 2027
Relevant Coursework: Computer Systems, Analysis of Algorithms, Concepts of Programming Languages, Database Systems, Geometric Algorithms, Competitive Programming II, Introduction to Artificial Intelligence, Software Engineering